

Mid-Position Adjustment

1. API Introduction

1. The API to open or close the bus servo torque is: **Arm_serial_set_torque(enable)**

Parameter explanation:

enable: enable=0 closes the torque, the robotic arm can be manually adjusted by 转动 the servo angle, and does not receive level signals; enable=1 opens the torque, the servo receives level signals to maintain the current angle, and commands are required to change the angle.

Return value: None.

2. **Arm_serial_servo_write_offset_switch(id)**

Parameter explanation:

id: When id=0, clear all servo mid-position adjustments and restore to default; id=1~6 corresponds to the IDs of the six servos. After receiving this command, the underlying microcontroller will read the angle data of the servo with the corresponding ID. If the data is within a reasonable range, it will be saved. If it exceeds the range or the servo ID cannot be found, it will not be saved.

Return value: None.

3. **Arm_serial_servo_write_offset_state()**

Parameter explanation:

Return value: Returns the state of the mid-position offset setting. state=0 indicates the servo was not detected; state=1 indicates the mid-position was set successfully; state=2 indicates the setting exceeded the mid-position range value.

2. Code Content

Code path: /home/yahboom/Dofbot/2.sys_settings/Offset/offset.ipynb

This code cannot be executed all at once in order. It must be executed step by step to ensure proper adjustment. The result of the adjustment is to keep the robotic arm in an upright state.

Note: This program should only be used when mid-position adjustment is needed. Do not use it arbitrarily, otherwise incorrect mid-position will affect the gripping效果.

```
#!/usr/bin/env python3
#coding=utf-8
import time
from Arm_Lib import Arm_Device

# Create robotic arm object
Arm = Arm_Device()
time.sleep(.1)
```

```
#Center and upright the servo state
```

```
Arm.Arm_serial_servo_write6(90,90, 90, 90, 90, 180, 1000)  
time.sleep(2)
```

```
# Close torque, now you can manually adjust servo angle by hand  
Arm.Arm_serial_set_torque(0)
```

```
# After adjusting the angle of a specific servo, you can set the mid-position  
offset for that servo individually  
id = 6  
Arm.Arm_serial_servo_write_offset_switch(id)  
time.sleep(.1)  
state = Arm.Arm_serial_servo_write_offset_state()  
if state == 1:  
    print("set offset ok!")  
elif state == 2:  
    print("error! set offset overrun !")  
elif state == 0:  
    print("error! set offset error !")
```

```
# You can also set the mid-position offset for all servos (1-6) at once  
for i in range(6):  
    id = i + 1  
    Arm.Arm_serial_servo_write_offset_switch(id)  
    time.sleep(.1)  
    state = Arm.Arm_serial_servo_write_offset_state()  
    if state == 1:  
        print("id:%d set offset ok!" % id)  
    elif state == 2:  
        print("error!id:%d set offset overrun !" % id)  
    elif state == 0:  
        print("error!id:%d set offset error !" % id)
```

```
# After adjustment is complete, open the torque  
Arm.Arm_serial_set_torque(1)
```

```
# Clear all servo mid-position offset settings and restore to default state.  
# If you need to clear the mid-position offset for all servos, remove the #  
below and run this cell  
#Arm.Arm_serial_servo_write_offset_switch(0)
```

```
del Arm # Release the Arm object
```

